

1.5 Polar coordinates

Further Pure 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Polar coordinates use $x=r \cos \theta$, $y=r \sin \theta$. Negative r represents a point in the opposite direction to θ , so specify conventions carefully. For a polar curve, area between $\theta=a$ and b is $\frac{1}{2}\int r^2 d\theta$; intersections may require considering repeated representations of the same point.

Worked example: follow the reasoning

For $r=2$ at $\theta=\pi/3$, Cartesian coordinates are $x=1$, $y=\sqrt{3}$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Find area swept by polar curve $r=2$ from $\theta=0$ to $\theta=\pi/2$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$$A = \frac{1}{2} \int_0^{\pi/2} r^2 d\theta [1] = \frac{1}{2} \times 4 \times \pi/2 [1] = \pi \text{ square units [1].}$$

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.